

CLIENT DELIVERABLE

SAOS Architecture Overview

Systack Documentation

ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b
Success	Green	 #22c55e
Error	Red	 #ef4444
Accent highlights	Purple	 #8b5cf6

SAOS Architecture Overview

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Prepared for: Systack Engineering

Prepared by: SOL + ORACLE

Builder: SOL + ORACLE + ASSEMBLY

Source System: SAOS v2026.06

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Support: Internal

SAOS System Architecture

Overview

SAOS (Systack Agent Orchestration System) is a 10-agent AI fleet deployed on dedicated VPS instances, orchestrated via PostgreSQL task queues and accessible through Tailscale-secured networks.

1. Fleet Composition

10-Agent Canonical Fleet

TIER	AGENT	EMOJI	ROLE	CAPABILITIES
Execution	SOL	🦄	Orchestrator	orchestration, execution, synthesis
Execution	CODY	🖥️	Build Engine	coding, voice, streaming
Execution	ASSEMBLY	🔧	Deployment	n8n, workflows, credentials
Quality/Risk	VALI	✅	Validation	testing, validation, quality
Quality/Risk	PESSI	⚠️	Risk Analysis	security, validation, risk
Intelligence	ORACLE	🧠	Design/Architecture	design, research, planning
Intelligence	ATLAS	🗺️	Knowledge	memory, documentation
Engagement	CHATTY	💬	Communication	communication, onboarding, content
Engagement	GENI	🎨	Creative	image_gen, video_gen, creative
Compliance	JURIS	⚖️	Legal/Compliance	legal, regulatory oversight

System Loop (Canonical)

```
ORACLE → Design → CODY → Build → ASSEMBLY → Deploy → VALI → Validate → PESSI → Stress-test → SOL → Execution
```

2. Infrastructure

VPS Tiers

PLAN	VULTR TIER	VCPU	RAM	STORAGE	MONTHLY COST
Business	vhp-8c-16gb-amd	8	16GB	160GB NVMe	\$96
Enterprise	voc-g-8c-32gb-160s-amd	8	32GB	160GB NVMe	\$240

Network Architecture

```
Client Device (Tailscale)
  ↓
Tailscale Tailnet (Encrypted Mesh)
  ↓
VPS (Ubuntu 22.04 + Tailscale)
  ├── Port 8765: Fleet Dashboard API
  ├── Port 8766: Invoice Dashboard API
  └── Port 8768: Customer Portal API
```

```
└─ Port 18789: OpenClaw Gateway
└─ PostgreSQL (systack_memory)
```

Persistence

SERVICE	PORT	LAUNCHAGENT	STATUS
Fleet Dashboard	8765	net.systack.dashboard	✓ Running
Invoice Dashboard	8766	net.systack.invoice-dashboard	✓ Running
Customer Portal	8768	net.systack.customer-dashboard	✓ Running
Orchestrator	N/A	net.systack.orchestrator	✓ Running
Webhook Bridge	8767	net.systack.webhook-bridge	✓ Running

3. Data Flow

Customer Portal Data Flow

```
Dashboard HTML (index.html)
  ↓ fetch every 30s
Customer Portal API (api.py :8768)
  ↓
PostgreSQL (systack_memory)
  ├── saos_clients (account info)
  ├── agent_state (agent status)
  ├── task_queue (task history)
  └── message_bus (inter-agent comms)
```

Provisioning Pipeline

```
Stripe Checkout
  ↓
n8n Webhook (test/production)
  ↓
saos_provision_bridge.py (:8767)
  ↓
PostgreSQL task_queue (DEPLOY task)
  ↓
Orchestrator polls every 10s
  ↓
ASSEMBLY executes provision_vps.py
  ↓
Vultr API → VPS created
  ↓
Client email sent
```

4. Security Model

Network Security

- **Tailnet-only** — no public internet exposure

- **HTTPS** — TLS termination via Tailscale
- **mTLS** — Tailscale handles device authentication

Dashboard Security (Current)

- `?client_id=` query parameter for data scoping
- **TODO:** Add JWT/session-based authentication
- **TODO:** Add login page

Credential Management

- OAuth secrets: Never committed (see RULE 7)
- API keys: Stored in PostgreSQL, not files
- Tailscale keys: Rotated via admin console

5. Monitoring & Health

Health Checks

- `/api/portal/health` — Customer portal API
- `/api/portal/status` — Full fleet status
- Tailscale serve status — `tailscale serve status`

Logs

SERVICE	LOG LOCATION
Customer Portal	<code>Systack/content/saos/saos-data/logs/customer-dashboard.log</code>
Fleet Dashboard	<code>Systack/content/saos/saos-data/logs/dashboard.log</code>
Invoice Pipeline	<code>Systack/content/saos/saos-data/logs/invoice-pipeline.log</code>

6. Known Issues & TODO

High Priority

- ☐ Dashboard authentication (login + session tokens)
- ☐ Vultr API key integration for auto-provisioning
- ☐ Stripe webhook production test

Medium Priority

- ☐ Billing portal integration
- ☐ Agent performance metrics
- ☐ Cost tracking dashboard

Completed

- ☒ Customer portal static file serving
- ☒ Tailscale serve configuration
- ☒ LaunchAgent auto-restart
- ☒ 10-agent fleet documentation

Document History

VERSION	DATE	CHANGES
1.0	2026-06-17	Initial architecture
2.0	2026-06-23	Added customer portal, 10-agent fleet, Tailscale serve

Latest version: Internal wiki